

ABITATHA ROY

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Undergraduate Researcher in Data Science and Machine Learning, with experience in Ensemble Methods, Environmental ML and Healthcare Prediction. Interested in Large Language Models.

Education

B.Tech. (Information Technology and Mathematical Innovations)	9.0/10 (90.0%)
Cluster Innovation Centre, University of Delhi	Expected 2027
Coursework: Data Science, Calculus, Linear Algebra, Probability and Statistics, Algorithms	

Technical Skills

Languages: Python, R | *Tools:* Streamlit, Flask, MATLAB | *Libraries:* Scikit-learn, TensorFlow/Keras, Pandas, NumPy, XGBoost, LightGBM

Experience

Cluster Innovation Centre, University of Delhi	New Delhi, India
Research Intern – Supervisor: Prof. Shobha Bagai	09/2025 – 11/2025
- Developed a feature-selection framework for AQI forecasting using CPCB air-quality datasets spanning four Indian cities, achieving R^2 up to 0.96 with compact predictor subsets. - Benchmarked ensemble learning models (CatBoost, XGBoost, LightGBM, Random Forest, AdaBoost) and analysed feature redundancy across heterogeneous environmental datasets. - Conducted episode-based spatio-temporal analysis of Delhi air pollution across seven monitoring stations, covering stubble burning, Diwali emissions, winter inversion and dust events.	
Ministry of Statistics and Programme Implementation, Government of India	Kolkata, India
Research Intern	06/2026 – present
- Analysing paradata and enterprise-level survey responses from the ongoing ASISSE survey under the Enterprise Survey Division. - Proposed anomaly detection frameworks using statistical methods and unsupervised learning to identify atypical response patterns across enterprises. - Working with senior statistical officers toward integrating data science methods into official statistics workflows.	

Projects

STAIR-TB: A Source-Aware Lightweight Stacked Ensemble Framework for Predicting Tuberculosis Treatment Failure	
Tech Stack: Python, Pandas, Scikit-learn, XGBoost, TensorFlow/Keras	
<i>Supervisor: Dr. Harendra Pal Singh</i>	
- Designed and implemented a lightweight stacked ensemble combining XGBoost, MLP and Logistic Regression for tuberculosis treatment-failure prediction. - Implemented source-aware stratification, preprocessing, feature selection and model evaluation workflows. - Achieved 0.67 recall and 0.6245 PR-AUC on a highly imbalanced dataset (22.3% positive prevalence); manuscript under preparation.	
Cross-Dataset Performance Evaluation of Predictive Algorithms for Phishing Website Classification	
Tech Stack: Python, Scikit-learn, XGBoost, Pandas	
<i>Supervisor: Prof. Shobha Bagai</i>	
- Developed a standardized URL feature-extraction pipeline comprising 17 handcrafted features. - Applied mutual-information-based feature selection to identify an optimal 9-feature subset, achieving 84.55% cross-dataset accuracy using Random Forest. - Conducted cross-dataset evaluation (Mendeley → PhiUSIIL) to assess domain-shift effects on phishing detection performance; implemented dataset harmonization to improve transferability across datasets.	

Achievements

2nd Position – MathWorks CodeMatrix Sprint	NIT Delhi, India
Tech Stack: MATLAB, Statistics and Machine Learning Toolbox, Symbolic Math Toolbox	10/2025
- Designed and implemented a gamified fitness tracking approach utilizing real-time mobile sensor data. - Performed real-time data processing and applied ML to calculate and report on user kinematics and calories.	

Publications

Clicks, Chats and Connections: How AI is Redefining Consumer-Brand Relationship	IIM Jammu, India
International Marketing Conference 2026	07/2026
Applied PLS-SEM modelling and clustering analysis on consumer survey data to evaluate the effects of AI Intelligence and AI Emotional Intelligence on brand satisfaction, trust and loyalty in chatbot-mediated interactions.	

Certifications

IBM Data Science Professional Certificate – 11/12 courses completed	Coursera
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